

Sub THz Pollutant Sensing: Attainable Model Error and Calibration Requirements

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Abstract—Atmospheric sensing claims based on a small Cramér–Rao bound require an attainable estimator and a credible error model. We construct the efficient linear Gaussian estimator for a layered HITRAN absorption model with four pollutant targets and atmospheric nuisance components. Real Beijing records set concentration scenarios and a training-only reference atmosphere; attenuation remains simulated. On 202 probes above 5 dB modeled SNR, explicit estimation reproduces the predicted noise error. However, systematic calibration and physical mismatch impose separate limits. At 300 coherent pilots, CO has normalized noise RMSE 0.994, leaving only 0.000740 dB tolerance to arbitrary bounded per-probe bias. At 30,000 pilots, SO₂ has matched-model RMSE 0.582 but reaches 3.918 under a declared combined temperature, pressure and vertical-profile mismatch. We provide exact conditional bias and variance formulas, noise simulations, covariance sensitivity and explicit resource accounting. These results establish attainability only within the stated Gaussian model and quantify why receiver calibration remains necessary for a physical sensing claim.

Index Terms—Atmospheric sensing, calibration, generalized least squares, HITRAN, integrated sensing and communication.

I. QUESTION AND CONTRIBUTION

Shared communication and absorption sensing has a precedent for atmospheric water vapour [1]; multi-medium propagation models motivate treating molecular and particle attenuation jointly [2]. Ambient pollutant retrieval additionally requires separating a weak target spectrum from background uncertainty. A small local lower bound alone cannot establish an achievable measurement error.

We ask which error requirements follow when that missing estimator is supplied. Our contribution is a reproducible conditional analysis: an explicit nuisance-eliminating estimator, its attainable covariance, its sensitivity to deterministic spectral bias, and an experiment in which a physically perturbed forward model differs from the estimator model. Generalized least squares and Fisher information are established methods [14]; this work applies them to the declared spectroscopy and link rather than claiming a new estimation principle. Earlier probe-selection, satellite-column and regression studies are retained in the appendix as supporting diagnostics.

II. EVIDENCE AND FIXED PHYSICAL MODEL

The primary spectroscopy input is the archived 9,340-line HITRAN table, checked against HAPI using the same line records [5], [6]. Real UCI Beijing records provide measured surface concentrations and weather [8]. They do not provide vertical profiles or synchronized radio attenuation.

The chronological split and concentration-order exclusions are unchanged. All current atmospheric summaries use training rows only; both training quantile scenarios and reused test-period regressions are explicitly separate from independent field validation.

Four reported errors are normalized by fixed concentration magnitudes: CO 4,000, O₃ 100, SO₂ 40 and NO₂ 25 $\mu\text{g m}^{-3}$. These numbers originate in WHO guideline values [12]; they are magnitude references, not detection probabilities, instantaneous compliance thresholds or a common averaging-period specification. Simulated estimates are not treated as health measurements.

The 24-layer atmosphere extends from 0 to 12 km. Line intensities, partition sums, pressure and Doppler broadening enter a Voigt absorption calculation with physical number-density units. The source surface concentration is mapped to an exponential gas profile with 1.5 km scale height. Water uses a 2 km profile; PM uses 1 km and the previously declared exploratory optical parameters. These profiles are assumptions. Reference temperature, pressure and dew point are taken from the training-only protocol. The explicit equations, line exclusions and prior implementation checks appear in the appendix.

The fixed receiver model retains the original 256 conceptual frequencies from 60 to 400 GHz, 550 km altitude, 45° elevation, 23 dBm total transmitted power, 0.5/0.3 m transmit/receive apertures, 1 MHz per probe, 6 dB noise figure and 5 dB implementation loss. Only 202 probes with reference SNR at least 5 dB enter the present estimator. Their power is not redistributed after masking. Thus the current study does not silently improve the link when opaque probes are removed. The frequency grid is a multiband modeling device, not a demonstrated contiguous waveform.

III. ATTAINABLE ESTIMATION UNDER DECLARED NOISE

After subtracting the reference background, use

$$y = D\theta + N\beta + n, \quad n \sim \mathcal{N}(0, \Sigma). \quad (1)$$

Columns of D contain the four magnitude-normalized gas spectra. N contains an offset, reference background scale and both PM spectra. The covariance is fixed and known in this conditional model; SNR is evaluated at the reference background, not re-estimated from unknown target values. A parameter-dependent physical noise model would require a different likelihood and Fisher calculation.

For linear SNR γ_i , $c = (10/\ln 10)^2$ and N_p pilots, the two declared thermal terms are

$$v_i^{\text{power}} = c(1 + \gamma_i^{-1})^2/N_p, \quad (2)$$

$$v_i^{\text{coherent}} = 2c/(N_p\gamma_i). \quad (3)$$

The power approximation is motivated by the pilot model in [7]; neither observation law has been calibrated for this receiver. Residual covariance is added as $\Sigma = \text{diag}(v) + \sigma_r^2 C$. We test $C = I$ and $C_{ij} = (1 - \rho)1\{i = j\} + \rho e^{-|f_i - f_j|/10 \text{ GHz}}$ with $\rho = 0.9$. Correlation direction and nuisance alignment can increase or decrease information. An independent residual standard deviation is not a systematic calibration specification.

Let $LL^T = \Sigma$, $\tilde{D} = L^{-1}D$, and let Q be an orthonormal basis for $L^{-1}N$. Set $P = I - QQ^T$ and

$$H = (\tilde{D}^T P \tilde{D})^{-1} \tilde{D}^T P L^{-1}. \quad (4)$$

If $P\tilde{D}$ has full target rank, $HD = I$, $HN = 0$ and

$$\hat{\theta} = Hy, \quad \mathbb{E}\hat{\theta} = \theta, \quad V = H\Sigma H^T = (\tilde{D}^T P \tilde{D})^{-1}. \quad (5)$$

These identities follow by substitution and $P^2 = P = P^T$. They attain the efficient Gaussian information bound in this unconstrained linear model [14]. They do not guarantee a nonnegative estimate or transfer to a misspecified nonlinear receiver. Our implementation uses Cholesky whitening and an SVD solve, not explicit normal-matrix inversion. Normalized nuisance and target rank tests use tolerance 10^{-12} ; a target absorbed by nuisance is rejected rather than assigned a finite precision.

IV. SYSTEMATIC CALIBRATION AND FORWARD-MODEL BIAS

If the actual mean differs by b , the same estimator obeys exactly

$$\text{bias}(\hat{\theta}) = Hb, \quad \text{MSE}_j = V_{jj} + (h_j^T b)^2, \quad (6)$$

where h_j^T is row j of H . When the fitted covariance is wrong, replace V by $H\Sigma_{\text{true}}H^T$. This sandwich calculation is included in the covariance experiment.

For an arbitrary persistent spectral error $\|b\|_\infty \leq \epsilon$, the exact targetwise worst case is

$$\sup_b \text{MSE}_j = V_{jj} + \epsilon^2 \|h_j\|_1^2. \quad (7)$$

The upper bound follows from the triangle inequality and is attained by $b_i = \epsilon \text{sign}(h_{ji})$. Different targets may attain their maxima at different bias vectors. For normalized RMSE at most one, the allowable deterministic error is therefore

$$\epsilon_j = \frac{\sqrt{1 - V_{jj}}}{\|h_j\|_1}, \quad V_{jj} \leq 1. \quad (8)$$

No allowance exists if thermal/residual noise already exceeds the target. An offset or modeled background-scale bias lies in $\text{col}(N)$ and is removed; this worst case concerns arbitrary bias shapes outside that nuisance model. Equations (6)–(8) are self-derived conditional sensitivity results, not measured calibration bounds.

TABLE I
COHERENT MATCHED-MODEL ESTIMATION, ZERO RANDOM RESIDUAL. ERROR IS DIVIDED BY THE GAS REFERENCE MAGNITUDE. THE FINAL COLUMN BOUNDS ARBITRARY SYSTEMATIC ERROR PER PROBE.

N_p	Gas	Predicted	Simulated	ϵ_j (dB)
300	CO	0.994	0.981	0.000740
3,000	CO	0.314	0.310	0.006530
30,000	CO	0.099	0.098	0.006845
30,000	SO ₂	0.582	0.575	0.000703
30,000	O ₃	1.250	1.246	None
30,000	NO ₂	12.133	12.191	None

TABLE II
NORMALIZED RMSE AT 30,000 COHERENT PILOTS, ZERO RANDOM RESIDUAL AND MEDIAN CONCENTRATION SCENARIO. PERTURBATION AMPLITUDES ARE ASSUMPTIONS, NOT MEASURED RECEIVER ERRORS.

Actual forward model	CO	O ₃	SO ₂	NO ₂
Matched	0.099	1.250	0.582	12.133
$T + 5$ K	0.420	2.130	2.992	13.311
$p - 2$ kPa	0.102	1.503	1.162	12.434
1.2 km gas height	0.112	1.255	0.584	12.137
Combined	0.484	1.472	3.918	14.745

V. FROZEN COMPUTATION AND RESULTS

The grid contains two likelihoods, four pilot counts (30, 300, 3,000, 30,000), three independent marginal residual scales (0, 0.001, 0.01 dB) and two covariance shapes. For each configuration, 10,000 new Gaussian observation-noise draws validate the actual linear estimator against its predicted covariance. Common standard-normal draws pair the configurations; they are not independent replications of physical data. We also evaluate deterministic physical mismatch at componentwise training concentration medians and 95th percentiles. These quantile vectors are declared scenarios, not necessarily jointly observed atmospheric states.

Across all configurations the maximum empirical-to-predicted RMSE discrepancy is 1.57 percent. This is a simulation consistency check, not an uncertainty interval for receiver performance. Earlier independent-noise calculations allowed CO at 300 pilots only below 0.00490 dB random residual. Under zero random residual, the separate worst-case systematic allowance is only 0.000740 dB. These numbers answer different questions and cannot be substituted for one another.

The mismatch experiment holds the estimator and its nominal covariance fixed. The true forward mean uses $T + 5$ K, $p - 2$ kPa, gas height 1.2 km, or all three together, including their background change. Temperature and pressure are not supplied to the estimator as corrected values. Noise covariance remains the declared nominal one so that this experiment isolates mean-model error; it is not a complete altered-link simulation. The exact bias formula evaluates the result without Monte Carlo uncertainty.

Matched-model SO₂ falls below its reference error, but a 5 K temperature mismatch alone increases its ratio from 0.582 to 2.992; the combined mismatch gives 3.918. More pilots reduce noise but do not remove this persistent model bias.

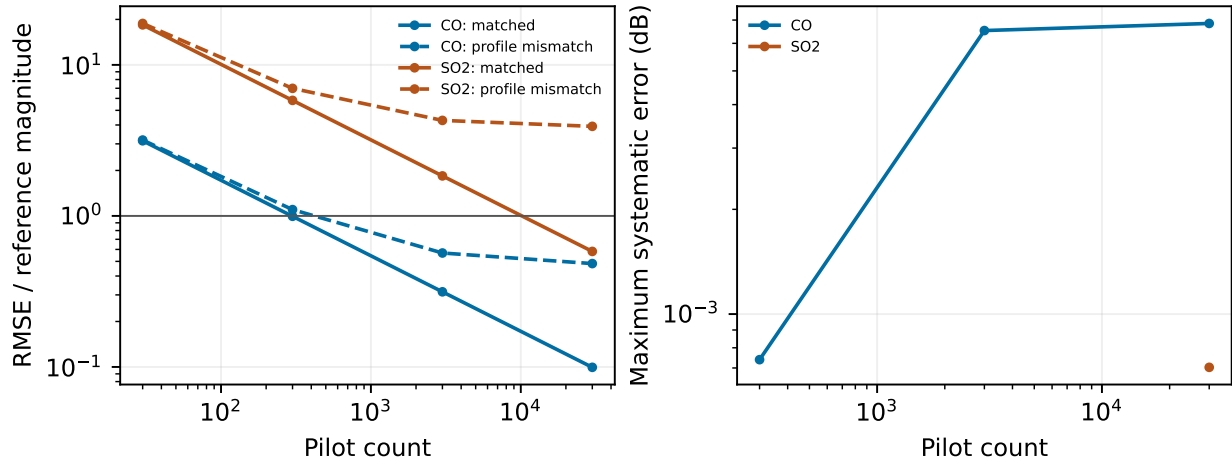


Fig. 1. Coherent estimation with zero random residual. Left: attainable matched-model error and combined physical mean mismatch at median concentrations. Right: exact targetwise allowance for arbitrary bounded systematic spectral error. Missing points already fail the noise target.

CO retains more margin at 30,000 pilots, while its 300-pilot result is fragile because its noise error already nearly reaches the target. Neither the chosen perturbations nor the worst-case spectral envelope constitute a calibrated uncertainty set for a real receiver.

VI. RESOURCE ACCOUNTING AND VALIDATION BOUNDARY

For a declared pilot duration T_s , the idealized simultaneous-probe observation duration is $N_p T_s$ and total transmitted energy is $P_{\text{total}} N_p T_s$. At the illustrative $T_s = 1 \mu\text{s}$ and 23 dBm, 300 pilots require 0.3 ms and 0.0599 mJ; 30,000 require 30 ms and 5.99 mJ. Retuning, coherent synchronization, guards and frequency-window access are excluded. The broad grid does not demonstrate that these probes can be acquired simultaneously with one front end. The larger budgets are therefore explicit resource scenarios, not free information gains.

A physical validation must supply paired attenuation and concentration/column truth, independent calibration recordings that estimate covariance and persistent bias across frequency and time, measured atmospheric profiles or justified uncertainty bounds, and a realizable frequency/power/timing allocation. The calibration data must be separated from the final evaluation period. A future acceptance criterion must specify the concentration averaging period and required estimation error before that evaluation. These are missing inputs, not completed experiments.

VII. CONCLUSION

An explicit efficient estimator now closes the gap between a local Gaussian lower bound and achievable error within the declared linear model. It does not close the physical calibration gap. Systematic bias has a different scaling from independent noise, and physically motivated mean perturbations can destroy the apparent SO₂ margin even at high pilot counts. The supported result is a conditional estimation

and calibration analysis using externally sourced spectroscopy and concentration scenarios. Successful atmospheric retrieval remains unestablished.

REPRODUCIBILITY

The new estimator, four analytical regression tests, frozen grid, observation-noise seed, coefficient arrays, physical mismatch scenarios, software, CPU timing, sampled memory and hashes are stored under `results/revision_0908`. The first confounding test exposed a floating-point residual rank issue; the implementation now compares projected target size with its preprojection size before normalization. This retained correction prevents a nuisance-confounded target from appearing identifiable. All primary tables are generated from the recorded CSV files.

APPENDIX A

SUPPORTING PHYSICS AND EARLIER EXPERIMENTS

The following diagnostics document earlier questions and parameter conventions. Their full-period reference atmosphere, 256-probe low-SNR extrapolation and historical comparisons do not replace the training-only, 202-probe primary protocol above. Old results are retained to disclose their limitations and the evidence motivating the current question.

A. Related Work and Scope

Aliaga *et al.* integrated differential absorption radar with beyond 100 GHz satellite communications and evaluated communication and water vapour sensing tradeoffs using atmospheric models [1]. Their work supports the shared sensing and communication premise, but it neither uses the UCI pollutants nor demonstrates their retrieval. Yang *et al.* model molecular and particulate attenuation across space air ground media and compare gaseous attenuation with ITU recommendations [2]. Their decomposition motivates our layered gas and particle treatment, while also highlighting that particle composition and size cannot be replaced by an arbitrary spectral trend.

ITU-R P.676-13 describes gaseous attenuation and ITU-R P.835-7 supplies reference atmospheres [3], [4]. We use them as propagation context, not as a claim that our trace gas columns or exploratory PM parameters are ITU calibrated. HITRAN2024 was the current edition when the local line table was retrieved through HAPI in July 2026 [5]. HAPI supplies partition sums and a reference implementation check using the same spectroscopic records [6].

Dong *et al.* derive CRB and Bayesian CRB limits for Ku band rain sensing from satellite pilot attenuation [7]. Their pilot power variance model is methodologically relevant, but their rain coefficients and validation data are not transferred to sub THz pollutants. The 0.63 dB residual value used below is borrowed only as a scenario value. It is not calibrated for the present frequency range or hardware. The gap addressed here is thus an estimation limit study for multiple ambient pollutants using public surface records and line data, with simulated rather than measured link attenuation.

B. Real Data, Provenance, and Quality

1) *Evidence Boundary*: Table III distinguishes external evidence from generated quantities. The Beijing data contain hourly measurements from 12 sites from March 2013 through February 2017 [8]. The ESA CCI records provide monthly total CO and tropospheric NO₂ columns at the same 39.5°N, 116.5°E grid cell from March through December 2013 [10], [11]. A CC BY Mendeley dataset supplies measured 0.9 to 1.3 THz aqueous protein spectra at 13 concentration levels across two proteins [9]. HITRAN parameters, UCI surface concentrations, satellite retrievals, and the laboratory spectra are real public data. The vertical shapes, LEO observations, receiver errors, and atmospheric estimator inputs are generated by the declared model. WHO values are health guideline concentration scales, not labels and not legal limits [12].

TABLE III
EVIDENCE AND ROLE IN THE STUDY

Item	Source	Count	Role
Spectra	HITRAN	9,340 lines	Real line positions, strengths, widths, shifts
Surface	UCI Beijing	383,585 rows	Real pollutant and meteorological scenarios
Column	ESA CCI	10 mo.	Real total CO and tropospheric NO ₂ retrievals
Lab THz	Mendeley	13 levels	Measured aqueous protein positive control
Health scale	WHO 2021	6 levels	Guideline comparison only
Sim. obs.	This study	20,000 rows	Simulated receiver observations

The processed HITRAN table spans 60 to 400 GHz and contains the main isotopologue for CO (15 lines), H₂O (37), O₂ (45), O₃ (1,301), NO₂ (1,536), and SO₂ (6,406). Its SHA256 is 7d063e4036da3d3e5... The cleaned UCI file SHA256 is 39d6ceee9d66824bcc... Full hashes, absolute input paths, package versions, and experiment assumptions are stored in the generated manifest.

2) *Air Quality Audit*: The original UCI record contains 420,768 station hour rows. Complete case processing retains 383,585 rows, or 91.163%. The retained table has no exact duplicate rows, no duplicate timestamp and station keys, no nonpositive target values, and no dew point above temperature. Station retention ranges from 86.704% to 94.616%.

The important failures are analytical rather than syntactic. In 17,642 rows (4.599%), PM₁₀ is below PM_{2.5}, so a physically nonnegative coarse mass cannot be formed. These rows are excluded from the estimator benchmark rather than silently clipped. Moreover, the real PM_{2.5}–PM₁₀ correlation is 0.8846, CO–PM_{2.5} is 0.7922, and NO₂–PM_{2.5} is 0.6707. A supervised model can therefore exploit population covariance even when a PM spectral component is undetectable. This is why regression accuracy alone is not accepted as evidence of direct spectral identification.

The complete case concentration fifth percentile, median, and ninety fifth percentile are respectively 200, 900, and 3,500 $\mu\text{g m}^{-3}$ for CO; 2, 45, and 177 $\mu\text{g m}^{-3}$ for O₃; 2, 7, and 59 $\mu\text{g m}^{-3}$ for SO₂; 8, 43, and 117 $\mu\text{g m}^{-3}$ for NO₂; 6, 55, and 242 $\mu\text{g m}^{-3}$ for PM_{2.5}; and 11, 82, and 279 $\mu\text{g m}^{-3}$ for PM₁₀. These records define realistic surface scenarios, not measured atmospheric columns.

3) *Satellite Column Audit*: The column experiment uses ten monthly files for each gas from March through December 2013. All 20 official objects downloaded anonymously, passed the declared NetCDF coordinate and variable checks, and retained valid values at the Beijing cell. CO source flags are 3 in every month and NO₂ quality flags are 1. The CO median is 3.317×10^{18} molecule cm⁻², its sample fifth to ninety fifth quantile span is 1.361×10^{18} , and the median reported uncertainty is 1.411×10^{17} . For tropospheric NO₂, the corresponding values are 1.304×10^{16} , 1.863×10^{16} , and 1.657×10^{15} molecule cm⁻². With only ten samples, the spans describe these available months rather than a stable population

distribution. The products are coarse satellite retrievals with instrument and prior dependence; they are closer to path quantities than station concentrations, but they are not direct atmospheric truth.

C. Layered Atmospheric Forward Model

1) *Line by Line Gas Absorption*: Let $\tilde{\nu}_k$ denote wavenumber and let $T_0 = 296$ K. The HITRAN reference intensity $S_\ell(T_0)$ is adjusted in layer j as

$$S_\ell(T_j) = S_\ell(T_0) \frac{Q(T_0)}{Q(T_j)} \exp \left[-c_2 E''_\ell \left(\frac{1}{T_j} - \frac{1}{T_0} \right) \right] \times \frac{1 - \exp(-c_2 \tilde{\nu}_\ell / T_j)}{1 - \exp(-c_2 \tilde{\nu}_\ell / T_0)}, \quad (9)$$

where $Q(T)$ is the HAPI partition sum and c_2 is the second radiation constant. Air pressure shift, air broadened Lorentz half width, and Doppler standard deviation are evaluated at each layer. The normalized Voigt function V_ℓ then gives

$$\sigma_{gjk} = \sum_{\ell \in g} S_\ell(T_j) V_\ell(\tilde{\nu}_k; T_j, p_j) \quad [\text{cm}^2 \text{ molecule}^{-1}]. \quad (10)$$

The lower atmosphere is divided into 24 equal thickness layers from 0 to 12 km. Temperature follows a standard lapse profile and pressure follows the corresponding barometric relation. Surface values are the medians of all retained UCI records: $T_s = 287.55$ K, $p_s = 101.04$ kPa, and dew point 2.9°C . This full period atmosphere is a declared transductive reference scenario, not a training only estimate. For target gas g , a measured surface mass concentration c_g is mapped to

$$n_g(z) = c_g \frac{N_A 10^{-12}}{M_g} \exp(-z/H_g) \quad [\text{molecule cm}^{-3}], \quad (11)$$

where c_g is in $\mu\text{g m}^{-3}$, M_g is in g mol^{-1} , and the assumed pollutant scale height is $H_g = 1.5$ km. The zenith attenuation design coefficient per unit surface concentration is

$$d_{kg}^{\text{zen}} = \frac{10}{\ln 10} \sum_j \sigma_{gjk} \frac{N_A 10^{-12}}{M_g} \exp(-z_j/H_g) \Delta z_j. \quad (12)$$

Here Δz_j is expressed in centimetres. H_2O number density is obtained from dew point and assigned a 2 km scale height; O_2 uses a dry air mole fraction of 0.20946. Their integrated attenuation forms the background vector $\mathbf{b}$. Target gas and PM scale heights remain assumptions, not measured UCI profiles.

For an elevation e , the reference calculation applies the plane parallel factor

$$d_{kg}(e) = \frac{d_{kg}^{\text{zen}}}{\sin e}, \quad b_k(e) = \frac{b_k^{\text{zen}}}{\sin e}. \quad (13)$$

This is adequate for a controlled sensitivity study but not an Earth curved ray tracing claim.

At the UCI median surface state, direct comparison with HAPI 1.3.0.0 gives peak cross section relative errors from

4.44×10^{-8} to 6.38×10^{-7} across the six molecules. Normalized RMSE over the active grid is below 1.58×10^{-7} for every molecule. This validates the implemented Voigt arithmetic against HAPI; it does not validate the assumed atmospheric profiles against measured columns.

2) *Native Column Design*: The satellite experiment does not convert columns into surface concentrations. For target g , let N_g^{col} be the retrieved column in molecule cm^{-2} and let nonnegative profile weights w_{gj} sum to one over the modeled layers. The zenith attenuation coefficient per unit column is

$$a_{kg} = \frac{10}{\ln 10} \sum_j \sigma_{gjk} w_{gj} \quad [\text{dB (molecule cm}^{-2})^{-1}], \quad (14)$$

so the zenith target contribution is $a_{kg} N_g^{\text{col}}$. At elevation e , the same plane parallel rule as (13) gives $a_{kg}(e) = a_{kg} / \sin e$. The implementation attaches an explicit domain to every column and rejects a total column supplied to a tropospheric parameter or the converse.

The column atmosphere uses 40 layers from 0 to 20 km at the March to December UCI medians $T_s = 290.85$ K, $p_s = 100.75$ kPa, and dew point 6.9°C . CO is a total column with an 8 km reference scale height, while NO_2 is tropospheric with a 1.5 km scale height. The model forces the reported total CO column into a normalized 0 to 20 km profile and therefore omits its true upper atmospheric distribution. The resulting line shape bias has unknown sign; CO scale heights of 4, 8, and 12 km test it only partially. Floors are compared with the real monthly column span and reported retrieval uncertainty, never with a surface WHO concentration scale.

3) *Exploratory PM Extinction*: For a spherical particle of radius r , complex refractive index m , wave number k_0 , and density ρ_p , the Rayleigh approximation uses [13]

$$\Phi = \frac{m^2 - 1}{m^2 + 2}, \quad (15)$$

$$C_{\text{abs}} = 4\pi k_0 r^3 |\text{Im } \Phi|, \quad C_{\text{sca}} = \frac{8\pi}{3} k_0^4 r^6 |\Phi|^2, \quad (16)$$

$$\kappa_{\text{ext}} = \frac{C_{\text{abs}} + C_{\text{sca}}}{4\pi r^3 \rho_p / 3}. \quad (17)$$

The fine mode assumes diameter 1 μm , density 1,500 kg m^{-3} , and $m = 1.50 + 0.01i$. The coarse mode uses 6 μm , 1,800 kg m^{-3} , and $m = 1.53 + 0.01i$. Both remain within the configured Rayleigh size parameter limit of 0.3 over the grid. Mass decays with an assumed 1 km scale height. In the benchmark, fine mass is $\text{PM}_{2.5}$ and coarse mass is $\max(\text{PM}_{10} - \text{PM}_{2.5}, 0)$. For a one parameter PM_{10} bound, the fine fraction is fixed to the median valid UCI ratio, 0.7681. These particle properties are exploratory, not calibrated Beijing aerosol optics.

D. LEO Link and Pilot Observation

1) *Reference Link*: The conceptual probe grid contains 256 equally spaced frequencies from 60 to 400 GHz, a spacing of 1.333 GHz. It samples possible multiband sensing locations;

TABLE IV
REFERENCE LINK AND OBSERVATION ASSUMPTIONS

Parameter	Value	Parameter	Value
Altitude	550 km	Elevation	45°
Slant range	749.1 km	Total P_t	23 dBm
Tx aperture	0.5 m	Rx aperture	0.3 m
Aperture efficiency	0.65	Active probes	256
Bandwidth/probe	1 MHz	Noise figure	6 dB
Receiver T	290 K	Other loss	5 dB
Pilot symbols	30	Residual std.	0.63 dB

it is not a proposed 340 GHz wide waveform. The received power on probe k is

$$P_{r,k} = P_{t,k} + G_t(f_k) + G_r(f_k) - L_{FS,k} - b_k(e) - L_{impl}, \quad (18)$$

where the total transmit power is divided equally across active probes, aperture gain is $\eta(\pi D/\lambda)^2$, and thermal noise is kTB plus receiver noise figure. Table IV records the reference assumptions.

At 45°, median background limited SNR is 14.97 dB, the fifth to ninety fifth percentile interval is $[-70.80, 20.64]$ dB, and 206 of 256 probes exceed 5 dB. Very opaque probes receive negligible Fisher weight.

2) *Pilot Attenuation Likelihood*: Following the pilot power argument in [7], the background referenced attenuation observation is

$$\hat{A}_k = \mathbf{d}_k^T \boldsymbol{\theta} + \beta_0 + \beta_b b_k + n_k, \quad (19)$$

where $\boldsymbol{\theta}$ is the target parameter vector, either surface concentration or native column amount, β_0 is an unknown offset, and β_b permits background amplitude mismatch. The independent Gaussian approximation uses

$$\text{var}(n_k) = \frac{c_0}{N_p} \left(1 + \frac{1}{\gamma_k}\right)^2 + \sigma_r^2, \quad c_0 = \left(\frac{10}{\ln 10}\right)^2. \quad (20)$$

Here γ_k is linear SNR, N_p is pilot count, $\sigma_k^2 = \text{var}(n_k)$, and $\sigma_r = 0.63$ dB is an assumed independent residual per probe. Independence is a declared scenario: real gain and atmospheric calibration errors can be correlated across frequency. At fixed marginal variance, correlation has no universal direction of effect after nuisance elimination, as the new covariance sensitivity demonstrates. Equation (20) is derived for moderate to high SNR. We extrapolate it only to assign very large variance, hence nearly zero weight, to opaque probes.

As an observation model sensitivity, a deterministic coherently averaged complex channel gives the high SNR delta method term $2c_0/(N_p\gamma_k)$ before adding the same residual variance. Unlike (20), it has no pilot power high SNR floor. Neither likelihood has been calibrated against a sub THz receiver; both remain declared observation scenarios.

E. Fisher Bounds and Evaluation Protocol

1) *Nuisance Aware CRB*: For the surface gas result, let $\mathbf{D}$ contain the four gas design columns and $\mathbf{N}$ contain normalized

nuisance columns for offset, $\text{H}_2\text{O}/\text{O}_2$ background, fine PM, and coarse PM. With $\mathbf{X} = [\mathbf{D} \ \mathbf{N}]$ and $\mathbf{W} = \text{diag}(1/\sigma_k^2)$, the Fisher information is [14]

$$\mathbf{J} = \mathbf{X}^T \mathbf{W} \mathbf{X}. \quad (21)$$

Eliminating nuisance parameters gives efficient gas information

$$\mathbf{J}_{\text{eff}} = \mathbf{J}_{gg} - \mathbf{J}_{gn} \mathbf{J}_{nn}^{-1} \mathbf{J}_{ng}, \quad (22)$$

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{J}_{\text{eff}}^{-1}.$$

The reference gas block has rank four and condition number 775.4. We define the one sigma floor for target g as $\sqrt{[\mathbf{J}_{\text{eff}}^{-1}]_{gg}}$ and also report three times this value. These are local unbiased estimator bounds conditional on all declared physics.

Joint PM retrieval is not reported as identifiable. The fine and coarse design columns have correlation 0.999999995, and the smallest to largest singular value ratio of the concentration scaled six column design is 2.61×10^{-7} . Instead, we show two optimistic one dimensional bounds: fine PM alone and fixed composition total PM₁₀. These are best case references, not evidence that both PM fractions can be recovered.

2) *Probe Selection and Power Allocation*: For a selected set $\mathcal{S}$, the regularized design objective is [15]

$$\Phi(\mathcal{S}) = \log \det \left(\lambda \mathbf{I} + \sum_{k \in \mathcal{S}} \tilde{\mathbf{x}}_k \tilde{\mathbf{x}}_k^T \right), \quad (23)$$

where $\tilde{\mathbf{x}}_k$ is the transposed design row after variance whitening, weighted column normalization, and a common numerical rescaling. Greedy selection maximizes the incremental value of (23). The ridge is 10^{-9} times the mean candidate information per parameter. Column normalization prevents units alone from determining the selection; the objective is not WHO weighted and does not first project out nuisance directions. For each requested tone count, σ_k^2 is computed in advance using equal division of the final total power. The surface experiment selects from 1,024 candidates and the native column experiment from 512 candidates across 60 to 400 GHz.

Uniform and D optimal designs use the same total transmit power, divided equally among the active probes, and the link variance is recomputed for every tone count. For 32 selected probes, a separate guideline scaled allocator maximizes $\log \det(\mathbf{J}_{\text{eff}})$ under the exact variance model. It scales the four gas parameters by their WHO concentrations, constrains power fractions to sum to one, and gives each probe at least 10^{-3} of its equal power fraction. This separates frequency placement from power redistribution. Increasing the nominal probe count also increases occupied bandwidth and the number of assumed independent residual samples in this experiment; it is therefore a design comparison, not a hardware scaling law.

3) *H₂O Local Control*: The H₂O control parameterizes an assumed 2 km water column by surface dew point. A centered finite difference with endpoints $\pm 0.25^\circ\text{C}$ around the training only median produces the local attenuation derivative on 512

candidate frequencies. The default D optimal design contains this derivative, an offset, four pollutant gas columns, and two PM columns. The strict design also contains the $\text{H}_2\text{O}/\text{O}_2$ background scale. Both use the same fixed total power and variance rule as the pollutant experiment. Bounds are evaluated for target only, offset plus pollutant nuisance, and the same set plus background scale. Endpoint forward evaluations check local nonlinearity, while paired training states at low and high humidity test transfer away from the reference temperature and pressure.

4) *Chronological Empirical Protocol*: After excluding the 17,642 PM ordering failures, 20,000 rows are selected deterministically over the sorted station hour table. A single median atmosphere generates attenuation from the real concentration vector, and Gaussian noise follows (20). Complete timestamp groups form 60% training, 20% validation, and 20% final test periods: 12,000, 4,000, and 4,000 rows. Training ends at 2015-09-10 01:00 and validation at 2016-06-03 04:00. The same 12 stations remain represented, so this tests temporal transfer but not unseen station or city transfer.

The training period mean is the data baseline. Ridge standardizes observations using training data only; α is selected only by validation normalized RMSE from 10^{-2} through 10^8 ; and the test set is evaluated once in the original benchmark. Per target RMSE is normalized by its training fifth to ninety fifth percentile range. Later test comparisons are diagnostic because follow up model exploration reused this period. Physical WLS uses the attenuation design directly without a label prior. An oracle joint WLS uses all gas and PM columns. A separate gas only test removes the true simulated PM contribution before inversion, then introduces a deliberate inverse mismatch of +5 K, -20 hPa, and a 1.2 km rather than 1.5 km pollutant scale height.

A post hoc metric audit fixes these rows, periods, training denominators, and targets. It adds a training prior LMMSE reference and a paired row bootstrap. The LMMSE uses only the training label mean and covariance as a Gaussian prior, then computes the posterior mean under the exact forward design and diagonal modeled noise covariance. It is therefore prior assisted and distinct from both supervised Ridge and label free physical WLS. Spectral model stability is tested over ten receiver noise seeds while holding real labels, forward signals, geometry, and the split fixed. Multitask ElasticNet and targetwise histogram gradient boosting configurations are frozen before this stability check.

The historical advanced diagnostic freezes 216 combinations of training or context prior, likelihood, probe objective, tone count, power allocation, nuisance projection, and non-negative posterior estimation. Validation alone selects every reported case. A second causal task uses 189 calendar, weather, station, and pollutant features whose target offsets are at least one hour. Station network reconstruction masks all six current query pollutant values before donor aggregation. Single channel repair instead masks only target j and permits the other five current query channels. Sentinel mutation tests enforce both masking contracts. These are ground sensor tasks and are not

TABLE V
REFERENCE FLOORS AT 45°; PEAK DB USES UCI Q95

Target	Peak dB	$1\sigma \mu\text{g m}^{-3}$	WHO $\mu\text{g m}^{-3}$	$1\sigma/\text{WHO}$	$3\sigma/\text{WHO}$
CO	0.03186	83,015	4,000	20.8	62.3
O ₃	0.00484	28,099	100	281.0	843.0
SO ₂	0.01517	4,308	40	107.7	323.1
NO ₂	0.00113	64,427	25	2,577.1	7,731.2
PM _{2.5}	0.00012	1,513,315	15	100,888	302,663
PM ₁₀	0.00014	1,581,204	45	35,138	105,414

counted as THz inversion. The measured THz control fixes the 0.9 to 1.3 THz band without labels, averages five frequencies, and evaluates ordinary least squares by leaving one measured concentration level out; its metric is not directly comparable with the Beijing chronological split.

F. Results

1) *Signal Magnitude*: Fig. 2 shows excess attenuation at the UCI ninety fifth percentile concentration and 45° elevation. The strongest gas peak is only 0.03186 dB for CO. O₃, SO₂, and NO₂ peak at 0.00484, 0.01517, and 0.00113 dB. Exploratory PM peaks are approximately 1.2×10^{-4} dB. Every value is far below the assumed 0.63 dB residual component, while total pilot error is larger still because of the first term in (20).

The most informative individual probes agree with the line structure: CO information is concentrated near 345.33 GHz, O₃ near 358.67 GHz, SO₂ near 357.33 GHz, and NO₂ near 396 GHz, with additional NO₂ information in lower loss windows. Strong cross section alone is not sufficient because $\text{H}_2\text{O}/\text{O}_2$ background can also suppress SNR.

2) *Reference Detection Floors*: Table V gives the main result. All reference one sigma bounds exceed the corresponding WHO concentration scales. The closest gas is CO at 20.8 times; NO₂ is 2,577 times. Three sigma ratios are exactly threefold larger. The PM values are already optimistic one mode bounds; joint fine and coarse retrieval is practically nonidentifiable.

The comparison is intentionally only a scale test. WHO values are health guidelines averaged over 8 h for O₃ and 24 h for the other targets, while the CRB describes one simulated attenuation burst and an assumed vertical mapping. The ratios do not establish a regulatory limit of detection, temporal averaging performance, or compliance classification.

3) *Sensitivity and Optimistic Stress Test*: The elevation sweep has a sensing optimum at the lowest tested elevation. At 15°, the longer assumed pollutant path outweighs the lower link SNR, reducing one sigma gas ratios to 10.4 for CO, 135 for O₃, 71.9 for SO₂, and 1,294 for NO₂. The result is conditional on the plane parallel profile; it is not a general claim that the lowest elevation is operationally optimal.

Increasing pilots from 30 to 3,000 at 45° reduces the CO ratio from 20.8 to 12.8, after which the 0.63 dB residual dominates. Increasing total power from 23 to 33 dBm changes the CO ratio only to 20.4. Setting the residual to zero while retaining 30 pilots gives 16.4. These one factor results show

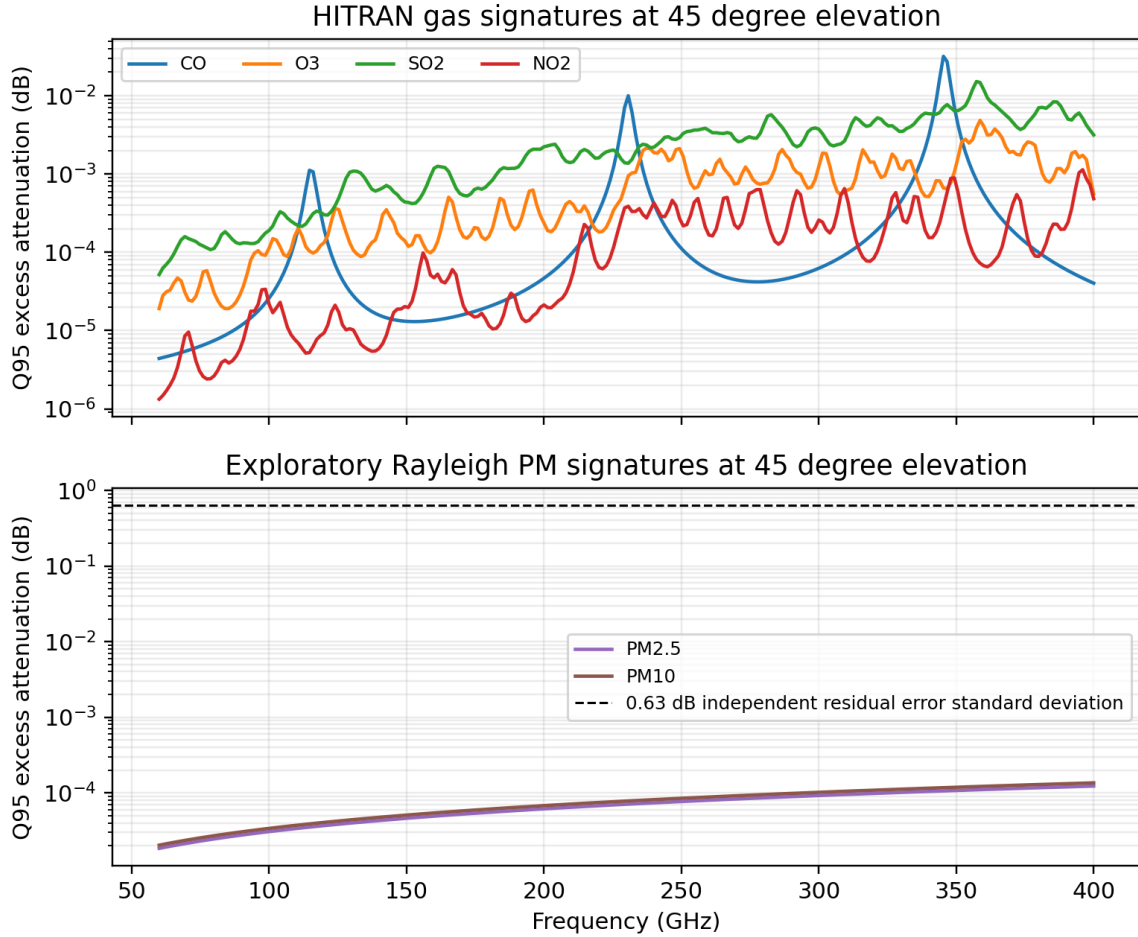


Fig. 2. Simulated excess attenuation for the complete case UCI ninety fifth percentile concentrations at 45° . Gas curves use the retrieved HITRAN line strengths. PM curves use the exploratory particle assumptions. The dashed line is only the assumed independent 0.63 dB residual component, not the total pilot error and not a measured instrument specification.

that the gap is not simply a low SNR problem: weak and overlapping sensitivities, opaque frequencies, pilot power estimation noise, and nuisance projection all matter.

We also combine every optimistic setting evaluated: 15° , 3,000 pilots, zero residual, 33 dBm total power, and all 256 probes. This is a stress test, not a proposed deployment. Its one sigma floor to guideline ratios are 0.649 for CO, 9.19 for O₃, 4.35 for SO₂, 88.2 for NO₂, 3,538 for PM_{2.5}, and 1,232 for PM₁₀. Only CO crosses the guideline concentration scale at one sigma, and its three sigma ratio is 1.95. The combined case therefore does not provide a robust multi pollutant detection success.

4) *Optimized Probes and H₂O Control*: Table VI applies greedy D optimal selection to the denser 1,024 point candidate grid. At 256 probes, the reference gas ratios improve by factors of 1.52 to 1.81 over the matched uniform selection, yet none approaches one. In the combined optimistic case, optimized CO reaches 0.353 at one sigma but remains 1.059 at three sigma. The other three gases remain above their guideline scales even at one sigma.

TABLE VI
GAS PROBE FLOORS DIVIDED BY WHO SCALE [12]. REF. VALUES USE 256 UNIFORM OR OPTIMIZED PROBES FROM THE 1,024 POINT CANDIDATE GRID; OPTIM. USES THE COMBINED STRESS TEST.

Target	Ref. unif.	Ref. opt.	Optim. opt. 1σ	Optim. opt. 3σ
CO	20.85	11.55	0.353	1.059
O ₃	280.85	184.67	5.925	17.776
SO ₂	107.64	63.94	2.594	7.783
NO ₂	2,579.94	1,664.56	55.676	167.028

Numerical power optimization under the stated variance model on the 32 selected probes adds only 0.2% to 4.3% improvement over equal power D optimal selection. Changing the regularizer from 10^{-6} to 10^{-12} leaves the 256 selected frequencies and floors unchanged. For optimistic CO, a one sigma ratio of one would require the independent residual to be at most 0.223 dB; a three sigma ratio of one is unattainable in this scenario even at zero residual. Reference case gases remain above one even at zero residual.

H₂O provides a modeled strong species control. Around

TABLE VII
NATIVE COLUMN RATIOS WITH DECLARED NUISANCE [10], [11]. Δq IS
THE TEN MONTH SAMPLE QUANTILE SPAN; u IS MEDIAN REPORTED
UNCERTAINTY.

Target	Peak dB	Ref./ Δq	Optim./ Δq	Ref./ u
CO total	5.97×10^{-3}	82.89	2.696	799.7
NO ₂ trop.	6.54×10^{-5}	5,476.27	179.97	61,559

the training only median state of 4.0°C dew point, 16.0°C temperature, and 1008.7 hPa, the 256 probe default design gives a local dew point floor of 0.290°C with offset, pollutant gas, and PM nuisance columns. When a background scale nuisance is added after that selection, the floor degrades to 2.541°C. A strict design that includes this nuisance during selection reduces the floor to 0.848°C, or 2.543°C at three sigma. Across pseudoinverse cutoffs from 10^{-10} to 10^{-15} the strict floor changes by less than 0.2%. Fixed temperature and pressure nonlinear checks remain within 8.2% relative error at the strict three sigma scale. Transfer to paired low and high humidity meteorological states produces 35% to 97% relative error, so this is a local information control rather than a global dew point estimator or measured retrieval.

5) *Real Satellite Column Stress Test*: Across the ten available months, the CO sample fifth to ninety fifth quantile span produces only 0.00597 dB peak zenith attenuation, while the tropospheric NO₂ span produces 6.54×10^{-5} dB. Table VII compares optimized local floors with each observed span Δq and the median retrieval uncertainty u . Even without nuisance parameters, the reference floor ratios are 76.96 for CO and 2,772.99 for NO₂. Declared nuisance columns increase them to 82.89 and 5,476.27. The combined optimistic scenario still gives one sigma ratios of 2.696 and 179.97, with three sigma ratios of 8.09 and 539.91.

Changing the CO scale height from 4 to 12 km changes its reference nuisance ratio from 107.1 to 74.7, which is material but does not change the conclusion. The native column result removes the surface concentration to column unit mismatch for these two species. It does not supply measured THz attenuation, and the ten monthly retrievals are too few and too coarse for an empirical sensing model.

6) *Empirical Estimation and Information Classes*: Validation originally selects Ridge $\alpha = 10^5$. On the chronological test period, its macro training Q05 to Q95 normalized RMSE is 0.34744 versus 0.34748 for the training period mean; mean R^2 is -0.08293 versus -0.08316 . Thus the apparent learned estimate is numerically a mean predictor, not evidence that the simulated spectrum carries usable target information.

The audit reproduces the Ridge score to 2.22×10^{-16} . The paired bootstrap difference, Ridge minus mean, has a 95% interval of $[-1.50 \times 10^{-4}, 7.64 \times 10^{-5}]$, so no difference is resolved under this conditional row bootstrap. The training prior LMMSE reference is 0.347435. Across the ten fixed noise realizations, Ridge is 0.347476 ± 0.000011 , where the latter value is the sample standard deviation across modeled noise seeds, not a confidence interval. Multitask ElasticNet

is 0.347741 and histogram gradient boosting is 0.347583. The frozen candidates beat Ridge on zero and one of ten seeds, respectively, so neither passes the predeclared eight seed stability rule. The alternative coherent CSI delta method sensitivity reaches only 0.345600 even after its independent residual is removed.

Previously explored forecasting, station reconstruction, single-channel repair and aqueous-protein controls are retained in the archived experiment notes. Their inputs and tasks differ from atmospheric THz inversion, so their scores are not used as sensing evidence here.

For the joint physical WLS, test RMSEs are $84,522 \mu\text{g m}^{-3}$ for CO, 27,845 for O₃, 4,223 for SO₂, and 58,383 for NO₂, close to the reference CRB scale in Table V. Approximately half of unconstrained WLS predictions are negative, as expected when noise dominates a zero centred linear inversion. Joint PM WLS is numerically unstable, with RMSEs of 6.36×10^9 and $1.45 \times 10^9 \mu\text{g m}^{-3}$ for PM_{2.5} and PM₁₀. The optimistic one parameter PM bounds must not be read as joint retrieval performance.

In the gas only oracle test, changing temperature, pressure, and scale height in the inverse model increases each gas RMSE by 31.5% to 32.6%. This controlled mismatch confirms that even the already poor matched inversion is optimistic. A constrained estimator could remove negative outputs, but it cannot create absent Fisher information.

7) *Training-only and calendar-block review replay*: The historical atmosphere used medians from the entire observation period. The review recomputes temperature, pressure, dew point, and the declared PM composition summary using training rows only. A mutation test verifies that changing future weather cannot alter the training atmosphere. The same frozen likelihood and Ridge validation grid are retained; the new run evaluates the selected model on the existing test period, which has already been inspected in earlier work. This is repaired computation on reused data, not an untouched confirmation set.

For the uncertainty calculation, all selected station rows inside each complete calendar block are resampled together. Seven-day and thirty-day partitions test sensitivity to the assumed dependence horizon. The estimand remains the paired difference in macro training-range normalized RMSE. The block intervals account for within-block station and temporal dependence, but cannot account for arbitrary dependence across blocks or variation between cities. The thirty-day analysis has only ten observed blocks and is particularly limited.

The training-only replay gives Ridge 0.34744429 and mean 0.34747886. The thirty-day paired difference interval is $[-1.0382 \times 10^{-4}, 3.2635 \times 10^{-5}]$; the seven-day interval also includes zero. The historical replay with calendar blocks likewise resolves no difference. The earlier row-bootstrap interval is retained above only as historical methodology and does not serve as the revised uncertainty claim. Raw predictions, test row indices, both block lengths, selected alpha, input hashes and training atmosphere are stored under `results/review`.

TABLE VIII
TRAINING-ATMOSPHERE ONE-SIGMA BOUNDS DIVIDED BY THE STATED
WHO REFERENCE CONCENTRATION SCALES. RANGES COVER THE
DECLARED COVARIANCE GRID ONLY.

Gas	Independent	Minimum	Maximum
CO	21.20	19.39	23.59
O3	289.41	279.95	352.09
SO2	111.17	103.92	125.72
NO2	2706.19	2489.12	2878.15

8) *Training-atmosphere correlated-noise sensitivity*: Independence is not a universal optimistic assumption when marginal noise variances are fixed. A correlated component aligned with an eliminated offset can be less damaging than independent errors of the same marginal variance, whereas a component aligned with a target spectral signature can be more damaging. For example, with two probes, target column $(1, -1)^T$, an unknown common offset, unit marginal variance and a common residual variance 0.4, the target variance bound is 0.3, compared with 0.5 for independent noise. A separate numerical test verifies this elementary counterexample.

The new physical sensitivity uses only the training-derived atmosphere of the repaired spectral replay: 290.35 K, 100,800 Pa and 5.4°C dew point. A check recomputes these quantities from training rows. The PM fine fraction is likewise training-derived. Let v_k denote the original total marginal variance and let $\sigma_r = 0.63$ dB remain the declared residual scenario. We keep every marginal variance unchanged and use

$$\Sigma = \text{diag}(v_k - \sigma_r^2) + \sigma_r^2 R, \quad R = (1 - r)I + r[e^{-|f_k - f_l|/\ell}]_{kl},$$

with $r \in \{0, 0.5, 0.9\}$ and $\ell \in \{1, 10, 100\}$ GHz. All nine covariance scenarios are frozen and retained. Whitening by the full covariance precedes nuisance elimination. We include an offset, background and both PM nuisance directions with a declared 10^{-12} relative SVD cutoff after column normalization; the nuisance rank is four in every case.

The fixed-marginal covariance changes can raise or lower the bound; for CO the range is 19.39–23.59 times the reference concentration scale, versus 21.20 under independence. No target approaches its reference scale in this grid. This reinforces the negative result only for the stated scenarios. It supplies neither a calibrated hardware covariance nor a worst-case envelope over all correlated errors. The reference atmosphere here differs from the historical full-period atmosphere in the earlier physical tables; those tables remain a declared historical scenario, not a prospective training-only calculation. Full configuration, input hashes and bounds are in `results/repair_correlated`.

9) *Conditional Receiver Requirements*: The preceding negative grids do not rule out all receiver/noise/resource combinations. We therefore compute explicit requirements rather than extrapolate a single borrowed residual value. The training-only atmosphere and reference link are retained. Both likelihoods now use the same 202 probes whose modeled SNR is at least 5 dB, avoiding low-SNR extrapolation in this comparison.

TABLE IX
CONDITIONAL ONE-SIGMA REQUIREMENTS. ZERO-RESIDUAL RATIO AND
MAXIMUM INDEPENDENT RESIDUAL (dB). ALL FOUR GASES AT THE
LARGEST COHERENT PILOT COUNT ARE SHOWN.

Model	N_p	Gas	Ratio	Max. σ_r
Power	30	CO	16.894	None
Power	300	CO	5.342	None
Power	3000	CO	1.689	None
Power	30000	CO	0.534	0.04055
Coherent	30	CO	3.144	None
Coherent	300	CO	0.994	0.00490
Coherent	3000	CO	0.314	0.04511
Coherent	30000	CO	0.099	0.04769
Coherent	30000	O3	1.250	None
Coherent	30000	SO2	0.582	0.00662
Coherent	30000	NO2	12.133	None

With $c = (10/\log 10)^2$, linear SNR γ_i , pilot count N_p and independent residual standard deviation σ_r , the declared alternatives are

$$v_i^{\text{power}} = \frac{c}{N_p} (1 + \gamma_i^{-1})^2 + \sigma_r^2, \quad (24)$$

$$v_i^{\text{coherent}} = \frac{2c}{N_p \gamma_i} + \sigma_r^2. \quad (25)$$

These are conditional likelihood scenarios, not measured receivers. Whitening and nuisance projection include an offset, background scale and both PM columns, with normalized nuisance SVD cutoff 10^{-12} . The four gas concentration columns are scaled by the same magnitude references as before. The resulting ratios are local one-sigma model bounds, not detection probabilities or compliance criteria.

The frozen grid uses 30, 300, 3,000 and 30,000 pilots and residual standard deviations 0, .001, .01, .1 and .63 dB. For each model and target, bisection reports the maximum independent residual in $[0, .63]$ dB for which the local floor is no larger than its reference scale. “None” means the target already fails at zero residual within that row’s resources.

Coherent CO crosses the one-sigma scale at 300 pilots only with a residual below .00490 dB; 3,000 pilots allow .04511 dB. At 30,000 pilots, coherent SO₂ also crosses, but requires residual below .00662 dB. O₃ and NO₂ still fail even at zero residual in the largest coherent row. Power-domain CO requires 30,000 pilots in this grid. Increasing pilot count increases observation time and total pilot energy at fixed per-pilot power; these rows are not equal-resource alternatives. The boundary is conditional on independent residuals and does not override the earlier covariance sensitivity.

The results in `results/resolution_noise_requirements` replace a blanket negative reading with explicit receiver requirements. They do not supply measured synchronized atmospheric attenuation, validate the physical profiles or turn a local bound into successful pollutant retrieval.

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